RoadGuard: An Explainable Deep Learning-Based Road Damage Assessment and Intelligent Maintenance Prioritization System

Problem Statement

Road infrastructure frequently develops potholes, cracks, surface deterioration, and other defects due to traffic, weather conditions, aging, and inadequate maintenance. Existing road inspection and complaint-based approaches often depend on manual identification and assessment, making it difficult to monitor large road networks consistently and prioritize critical repairs. Therefore, an intelligent system is required to automatically detect road damage, estimate its severity, identify its geographic location and responsible authority, and prioritize maintenance based on road-condition and risk factors.

Abstract

RoadGuard is an AI-based road infrastructure assessment and maintenance prioritization system designed to automate the detection, analysis, and management of road defects. The proposed system processes road images or videos captured using mobile devices or vehicle-mounted cameras and uses a deep learning-based object detection model to identify defects such as potholes, longitudinal cracks, transverse cracks, alligator cracks, and surface damage. The detected defects are analyzed to estimate their severity based on factors such as damage type, size, and defect density.

GPS coordinates are associated with each detected defect to identify the corresponding road segment and determine its road type and jurisdiction using a road-authority mapping database. A maintenance-priority model then generates a priority score based on damage severity, defect concentration, road importance, location risk, and repeated reports. The results are stored in a database and visualized through an interactive map and authority dashboard, allowing responsible departments to identify critical road segments and manage maintenance activities efficiently. The system supports both user-reported complaints and proactive image-based road inspection, enabling previously unreported road damage to be identified.

The proposed system extends existing deep-learning-based pavement distress detection approaches by integrating **severity assessment, geospatial mapping, authority identification, explainable predictions, and intelligent maintenance prioritization** into a unified workflow. The prototype will be evaluated using public road-damage datasets and locally collected road images using precision, recall, F1-score, mAP, and inference time. The system aims to provide a scalable and data-driven approach for improving road-condition monitoring and maintenance decision-making.

Keywords

Road Damage Detection, Pothole Detection, Crack Detection, Deep Learning, YOLO, Computer Vision, Explainable AI, Severity Estimation, GPS Localization, Geospatial Mapping, Road Condition Assessment, Maintenance Prioritization, Intelligent Infrastructure Monitoring